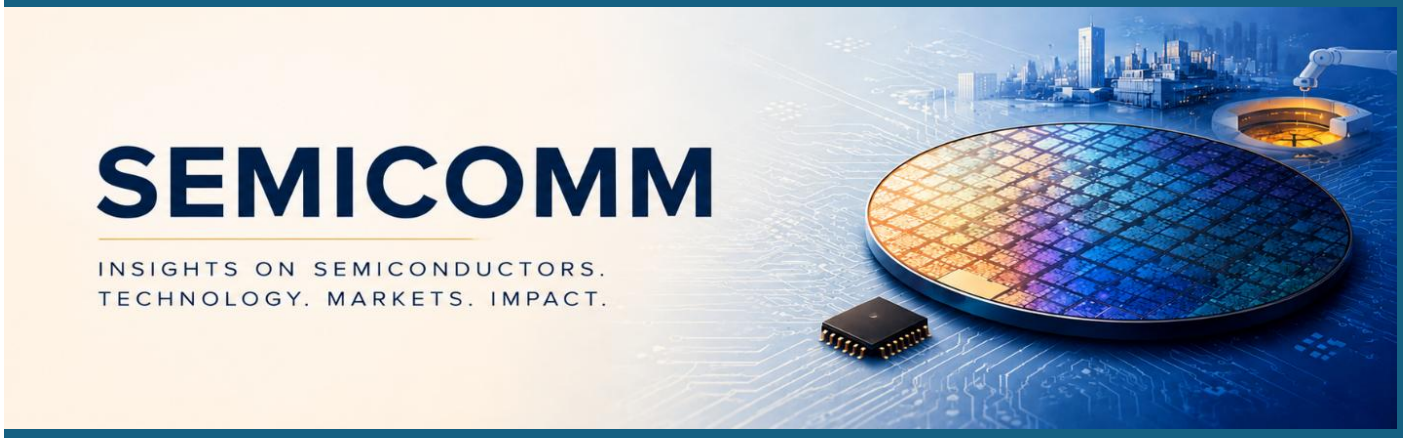
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The Semiconductor Fab Handbook

A Bottom-Up Guide to Building, Operating, and Profiting From a 28nm/45nm Semiconductor Fab

The Semiconductor Fab Handbook



Published by SemiComm (https://semicommhub.com/)

White paper for new semiconductor manufacturers, investors, materials suppliers, and ecosystem planners

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Executive Summary

Semiconductors have become one of the world's most strategically important industries, underpinning artificial intelligence, cloud computing, automotive electronics, industrial automation, communications infrastructure, and consumer devices. As governments and corporations invest billions of dollars into new manufacturing capacity, a fundamental question remains:

What does it take to build and operate a successful semiconductor fabrication facility?

Building a semiconductor fab is not simply a construction project. It is a long-cycle manufacturing capability project that combines process technology, equipment, materials, utilities, workforce development, yield learning, customer qualification, supply-chain resilience, and financial discipline.

For a new manufacturer, the central question is not only whether a fab can be built, but whether it can reach:

- Stable yield
- Qualified customers
- High utilization
- Reliable utilities
- Controlled materials
- Competitive wafer cost
- Sustainable profitability

A full-scale 300mm mature-node fab operating at 40,000-50,000 wafer starts per month (WSPM) requires significant capital, technology access, operational expertise, and customer demand. As a public benchmark, Tata Electronics' Dholera 300mm fab has been described as a 50,000 wafers-per-month facility spanning 28nm to 110nm technologies, with an announced investment of approximately US$11 billion.[1-3]

For most new entrants, a pure 28nm-first strategy is rarely the optimal path. A more realistic approach is to establish a 300mm **mature-node** platform, initially ramping more mature (90-45nm technologies before introducing 28nm. This progression allows the manufacturer to build stable process control, low defectivity, customer-qualified process design kits (PDKs), reliable utility infrastructure, and predictable yield performance.[4]

This white paper examines what it takes to build and operate a semiconductor manufacturing fab from the ground up. Through a structured end-to-end assessment (using a representative base case as below), we analyze the technology, infrastructure, equipment, materials, utilities, workforce, yield, supply-chain, customer, and financial requirements needed to transform a greenfield fab investment into a sustainable and profitable semiconductor manufacturing business.

Representative Base-Case Fab used for this study:

Reasons for these assumptions are explained in the next sections.

Metric	Base-Case Assumption
Wafer size	300mm
Installed capacity	40,000 WSPM
Target utilization	85%
Effective wafers/year	408,000 wafers
Technology path	45/40nm ramp → 28nm expansion
Cleanroom area	20,000-40,000 m^2
Employees	~2,500-3,000
CAPEX range	US$7B-10B (planning estimate)
Revenue potential	~US$1.0B/year at US$2,500/wafer
Yield target	>85% mature functional yield
Tool availability target	>90% for critical tools

Key takeaway:

A fab becomes successful only when the building, tools, process, materials, people, customers, and economics all work together. A cleanroom without qualified customers is a depreciation machine. A process without yield is not a business. A 28nm fab without high utilization is financially fragile.

1. The Technology-Market Decision

The first decision is not technical; it is commercial. Before selecting a process technology, a new manufacturer must determine which node offers the optimal balance between market demand, manufacturing complexity, capital intensity, customer requirements, and long-term competitive differentiation.

This section evaluates the relative attractiveness of key semiconductor nodes and identifies the factors that influence node selection for a new fab entrant, including process complexity, capital expenditure requirements, ecosystem maturity, customer demand, and long-term market potential.

1.1 Technology Node comparison

When discussing semiconductor manufacturing, attention is often focused on leading-edge technologies such as 3nm, 2nm, and future process nodes. However, the majority of semiconductor volume continues to be produced on mature and mid-mature technologies. Automotive electronics, industrial automation, communication infrastructure, consumer devices, display controllers, power management ICs, and many IoT products do not require the most advanced transistor geometries to remain competitive.

Node	Typical applications	Complexity	Capex	Market attractiveness
180nm	Automotive, industrial, analog, power	Low	Low	Mature
130nm	Microcontroller unit (MCU), analog, automotive	Low-medium	Low	Mature
90nm	MCU, connectivity, industrial	Medium	Medium	Mature
65nm	MCU, connectivity, display, embedded logic	Medium	Medium	Strong
45/40nm	MCU, automotive, connectivity, display, mixed-signal	Medium-High	High	Strong
28nm	Connectivity, display, automotive logic, edge AI, consumer SoC	High	High	Very strong
22nm FD-SOI	Automotive, IoT, RF, low-power logic	High	High	Niche/specialty

For a new semiconductor manufacturer, the critical question is therefore not: "What is the most advanced node available?" but rather: **"Which node provides the best balance between market demand, manufacturing complexity, capital investment, and long-term profitability?"**

Among mature and mid-mature technologies, **28nm** has emerged as one of the industry's most commercially successful process nodes, often referred to as the industry's "golden node", the reasons for which are discussed in the next section.

1.2 Why 28nm Became a Strategic "Golden Node"

28nm technology node occupies an unusual position in semiconductor manufacturing, as it is in between traditional mature technologies and advanced nodes. This node provides substantial improvements in transistor density, power efficiency, and integration capability while avoiding the extreme complexity and cost associated with FinFET and EUV-based manufacturing.[5]

The commercial success of 28nm can largely be attributed to six structural advantages:

- Strong balance between performance and manufacturing cost
- Broad applicability across multiple end markets
- Long commercial lifecycle
- Attractive return on invested capital
- Large and mature design ecosystem
- Extensive global manufacturing infrastructure

Unlike older mature nodes, 28nm delivers sufficient performance for complex digital, connectivity, display, and automotive applications. At the same time, it avoids many of the manufacturing challenges introduced by FinFET-based technologies.

Feature	28nm	20nm	16nm / 14nm and below
Transistor type	Mostly planar CMOS, often HKMG variants	Planar, more complex	FinFET
Lithography	DUV immersion / dry DUV	DUV multiple-patterning heavy	DUV + EUV or advanced multi-patterning
Manufacturing complexity	High but manageable	Higher	Very high
Design cost	High	Higher	Very high
Economic position	Strong mature-node sweet spot	Transitional	Advanced-node economics

From a manufacturing perspective, 28nm is particularly attractive because it remains a predominantly planar CMOS technology, allowing fabs to leverage mature **DUV lithography** infrastructure while achieving significantly higher integration levels than 45nm or 65nm processes.

1.3 Why choose even more mature nodes (e.g. 45/40nm) as a new manufacturer?

Advantages of 28nm

A manufacturer may choose 28nm because it offers:

- Higher transistor density
- Lower cost per function
- Better power-performance characteristics
- Smaller die size for complex designs
- Strong relevance for:
    - Connectivity ICs
    - Display controllers
    - Advanced MCUs
    - Edge AI processors
    - Consumer SoCs
    - Automotive control systems
- Higher potential wafer average selling price (ASP) compared with older mature nodes. Public estimates place 28nm 300mm wafer pricing around US$ 2,550 – US$ 3,450, with an average near US$ 3,000.[7]

The node continues to support high-volume production across consumer, industrial, networking, and automotive markets while remaining accessible to DUV-based manufacturing.

Advantages of 45/40nm

However, a new manufacturer may instead prioritize 45/40nm because it provides:

- Lower process complexity.
- Faster yield learning.
- Lower defect sensitivity.
- Simpler lithography requirements.
- Lower customer qualification risk.
- Strong automotive and industrial adoption.
- Good wafer ASP in the estimated range of US$1,800 – 2,500 per 300mm wafer.

For a first-time fab operator, these factors can significantly reduce execution risk during the early years of operation.

Strategic Recommendation

For most new semiconductor manufacturers, the optimal strategy is not to begin with high-volume 28nm production immediately. A more practical roadmap is:

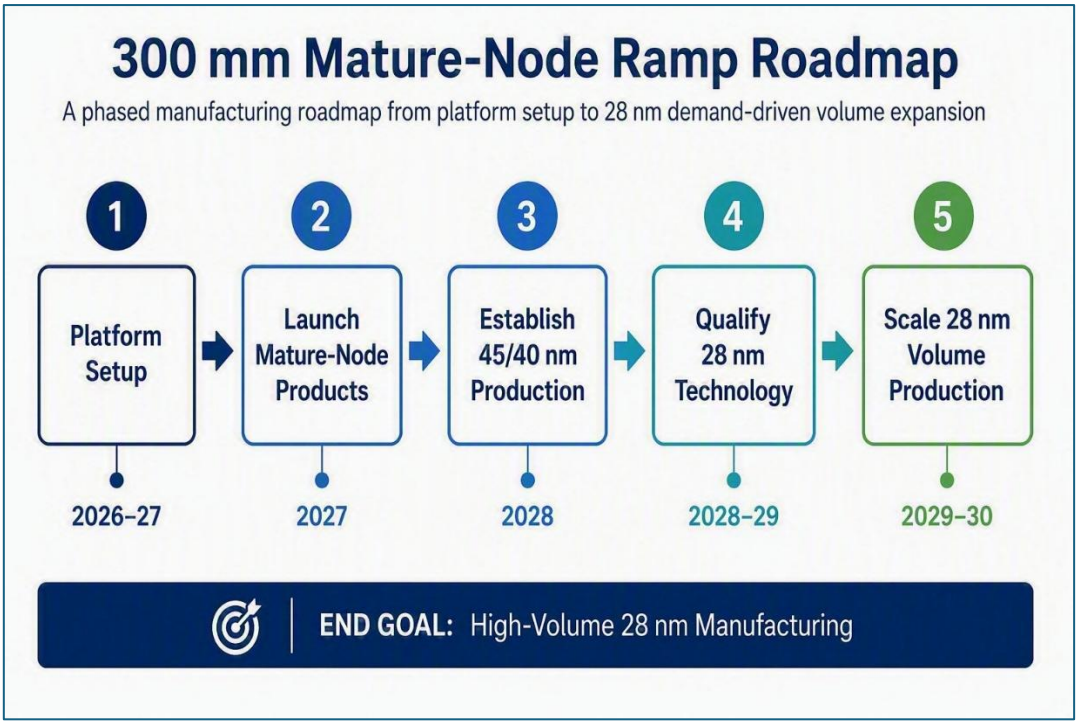


Figure 1: Simplified roadmap illustrating the phased evolution of a 300 mm fab from platform setup to high-volume 28 nm manufacturing.

This phased progression allows the organization to build the following capabilities before introducing the additional complexity associated with large-scale 28nm production:

- Process integration expertise.
- Yield-management capability.
- Metrology and inspection discipline.
- Manufacturing culture.
- Customer confidence.
- Supply-chain maturity.

Key takeaway:

Mature nodes remain the foundation of modern semiconductor manufacturing. Among them, 28nm represents one of the best balances ever achieved between performance, manufacturing complexity, ecosystem maturity, capital efficiency, and market demand, making it a highly attractive destination for new semiconductor manufacturers. However, a new manufacturer should usually build a 45/40nm-capable platform first and use it as the operational bridge to 28nm.

2. The Hypothetical Fab Definition

Before evaluating equipment, materials, workforce, or financial performance, the manufacturing strategy must be clearly defined. The selected technology node, capacity, product mix, and business model determine virtually every aspect of fab design, including:

- Cleanroom size
- Equipment requirements
- Utility demand
- Workforce requirements
- Process complexity
- Supply-chain requirements
- Capital expenditure
- Customer qualification requirements

For this white paper, a representative mature-node foundry is modelled to evaluate the technical and economic feasibility of a new semiconductor manufacturing operation.

***Important**: This is an analytical base case. Actual cost and ramp timing depend heavily on country, subsidies, labour cost, utility reliability, local ecosystem, equipment availability, technology partner, process complexity, and construction strategy.*

2.1 Base-case fab specification

Commercial Definition

Parameter	Base-Case Assumption
Fab Type	Greenfield 300mm Foundry
Business Model	Pure-play foundry
Technology Platform	45/40nm initial production, 28nm expansion
Wafer Size	300mm
Installed Capacity	40,000 WSPM
Utilization Target	85-90%
Ramp Period	5 years
Primary Markets	Automotive, Industrial, Connectivity, Display, MCU, Edge Computing
Packaging Strategy	External OSAT during initial ramp
Technology Source	Licensed platform or technology-transfer partner

Technology Platform Assumptions (45/40nm Initial Production with 28nm Expansion)

Parameter	Base-Case Assumption
Technology Platform	45/40nm initial production, expandable to 28nm
Transistor Architecture	Planar CMOS
Gate Stack	Poly-Si gate for 45/40nm; High-K Metal Gate (HKMG) for 28nm
Isolation Scheme	Shallow Trench Isolation (STI)
Source/Drain Formation	Ion implantation with advanced anneal integration
FEOL Complexity	High (45/40nm), Very High (28nm)
FEOL Process Steps	~200-300 (45/40nm), ~250-350 (28nm)
FEOL Mask Layers	~15-20 (45/40nm), ~18-25 (28nm)
Critical FEOL Lithography Layers	~8-12 (45/40nm), ~12-18 (28nm)
MOL Complexity	Contact and local interconnect formation
MOL Process Share	~10-15% of total process flow
BEOL Metal Layers	8-10 (45/40nm), 10-12 (28nm)
BEOL Process Share	~35-45% of total process flow
Lithography Platform	ArF Immersion (193nm), ArF Dry (193nm), KrF (248nm)
Lithography Burden	High (45/40nm), Very High (28nm)
Metrology Burden	High (45/40nm), Very High (28nm)
Total Mask Count	~35-45 (45/40nm), ~40-55 (28nm)
Total Process Steps	~450-650 (45/40nm), ~600-800 (28nm)
Fabrication Cycle Time	~50-80 days (45/40nm), ~60-90 days (28nm)
Design Enablement	Standard-cell libraries, SRAM compilers, I/O libraries
Process Design Kit (PDK)	Full customer-qualified PDK
Reliability Capability	Automotive and industrial qualification support
Defect Density Target	<0.1 defects/cm^2 at mature production
Target Functional Yield	85-95% at mature production
Process Integration Difficulty	High (45/40nm), Very High (28nm)
Technology Source	Licensed platform, JV partner, or technology-transfer partner

Physical Scale Assumptions

Parameter	Planning Value
Total site area	100-250 acres
Fab building area	40,000-80,000 m^2
Cleanroom area	20,000-40,000 m^2
Utility area	15,000-30,000 m^2
Warehouse area	5,000-15,000 m^2
Office/support area	5,000-10,000 m^2

Process Complexity Assumptions (Compiled)

Metric	45/40nm	28nm
FEOL Masks	15-20	18-25
MOL Masks	2-5	3-5
BEOL Masks	20-25	22-30
Total Masks	35-45	40-55
FEOL Steps	200-300	250-350
MOL Steps	50-80	70-100
BEOL Steps	200-270	280-350
Total Steps	450-650	600-800
Critical Lithography Layers	15-20	20-30
Fabrication Cycle Time	50-80 days	60-90 days
Lithography Burden	High	Very High
Metrology Burden	High	Very High
Process Integration Difficulty	High	Very High

Important: These are planning-level assumptions. Actual numbers depend on process architecture, product type, foundry platform, reliability options, and customer requirements.

Why This Configuration?

This configuration was selected because it represents the most realistic entry point for a new foundry operator:

- Large enough to achieve manufacturing scale
- Mature enough to reduce technology risk

- Advanced enough to attract connectivity, automotive, and display customers
- Does not require EUV infrastructure
- Supports long-term expansion into higher-value nodes

2.2 Manufacturing Targets

A fab should not be defined only by wafer starts. It should be defined by the products it intends to manufacture.

Target Product Portfolio

Product Type	Typical Process Node
Automotive MCU	40nm - 90nm
Industrial MCU	40nm - 130nm
Connectivity IC	28nm - 65nm
Display Driver IC	28nm - 55nm
Network Processor	28nm - 45nm
Edge AI Controller	28nm
Industrial SoC	28nm - 45nm
Security Controller	40nm - 90nm

Typical Die Characteristics

Metric	Assumption
Average Die Size	50-150 mm^2
Gross Dies / 300mm Wafer	~400-1300
Functional Yield Target	85-95%
Wafer Sort Yield	95-99%
Packaging Yield	98-99.5%
Final Test Yield	95-99%

These values are illustrative and product dependent.

A 50mm^2 die and a 150mm^2 die processed on the same node can have dramatically different economics because gross die count and defect sensitivity change with die area.

These assumptions will later drive:

- Revenue
- Material consumption
- Yield
- Cost per good die
- Packaging requirements
- Customer economics

2.3 Required site elements

A 40,000 WSPM 300mm fab requires more than a cleanroom. The manufacturer must plan for:

- Land
- Fab building
- Cleanroom
- Sub-fab
- Office and support buildings
- Utility buildings
- Chemical storage and distribution
- Specialty gas farms
- Bulk gas systems
- Wastewater treatment
- Electrical substation
- UPW plant
- Chillers
- HVAC systems
- Exhaust and abatement
- Warehouse
- Logistics docks
- Maintenance shops
- Training center
- Emergency response infrastructure
- EHS infrastructure
- Security infrastructure
- Data and manufacturing-control infrastructure

Representative FAB value-creation process flow:



Figure 2: Semiconductor fab value-creation flow, illustrating the progression from site and infrastructure readiness to qualified products, customers, and cash generation.

3. Fab Architecture and Process Flow

A fab converts incoming silicon (Si) wafers into finished wafers through repeated cycles of patterning, material addition, material removal, doping, planarization, cleaning, and measurement. Semiconductor fabrication involves many sequential process steps, including lithography, deposition, etch, implant, diffusion/anneal, CMP, cleaning, metrology, wafer sort, assembly, packaging, and testing.[9]

3.1 High-level process flow



Figure 3: High-level 300 mm semiconductor manufacturing flow from wafer input to customer-qualified mass production.

3.2 FEOL, MEOL, and BEOL split

Module	Purpose	Approximate Process Share
FEOL	Transistor formation	40-50%
MOL	Contact and local interconnect	10-15%
BEOL	Metal interconnect and passivation	35-45%

3.3 Major process modules

Module	Typical Equipment	Key Materials	Major Yield Risk	Cost Driver
Lithography	Scanner, coat/develop track	Resist, BARC, developer, solvents	CD variation, overlay, defects	Scanner/track utilization
Etch	Plasma etcher	CF_4, Cl_2, HBr, O_2, C_4F_8	Residue, profile control, selectivity	Chamber productivity
Deposition	CVD, ALD, PVD	Precursors, targets	Film defects, thickness variation	Tool throughput, precursor consumption
Implant	Ion implanter	Dopant sources/gases	Dose/energy variation, Vt variation	Implant utilization
Thermal	RTP, furnace	N_2, O_2, process gases	Activation issues, thermal excursions	Thermal budget / throughput
CMP	CMP polisher, post-CMP clean	Slurry, pads, conditioners	Scratches, dishing, erosion	Consumables + throughput
Cleans	Wet bench, single-wafer clean	HF, HCl, NH_4OH, H_2O_2, UPW	Particles, metallic contamination	Chemical + UPW consumption
Metrology	CD-SEM, OCD, film/overlay tools	Reference wafers, calibration standards	Measurement drift, tool matching	Sampling rate / throughput
Inspection	Optical inspection, e-beam review	Inspection recipes, calibration wafers	Defect escape, detection sensitivity	Throughput / sampling intensity

For every module, a fab manufacturer must define:

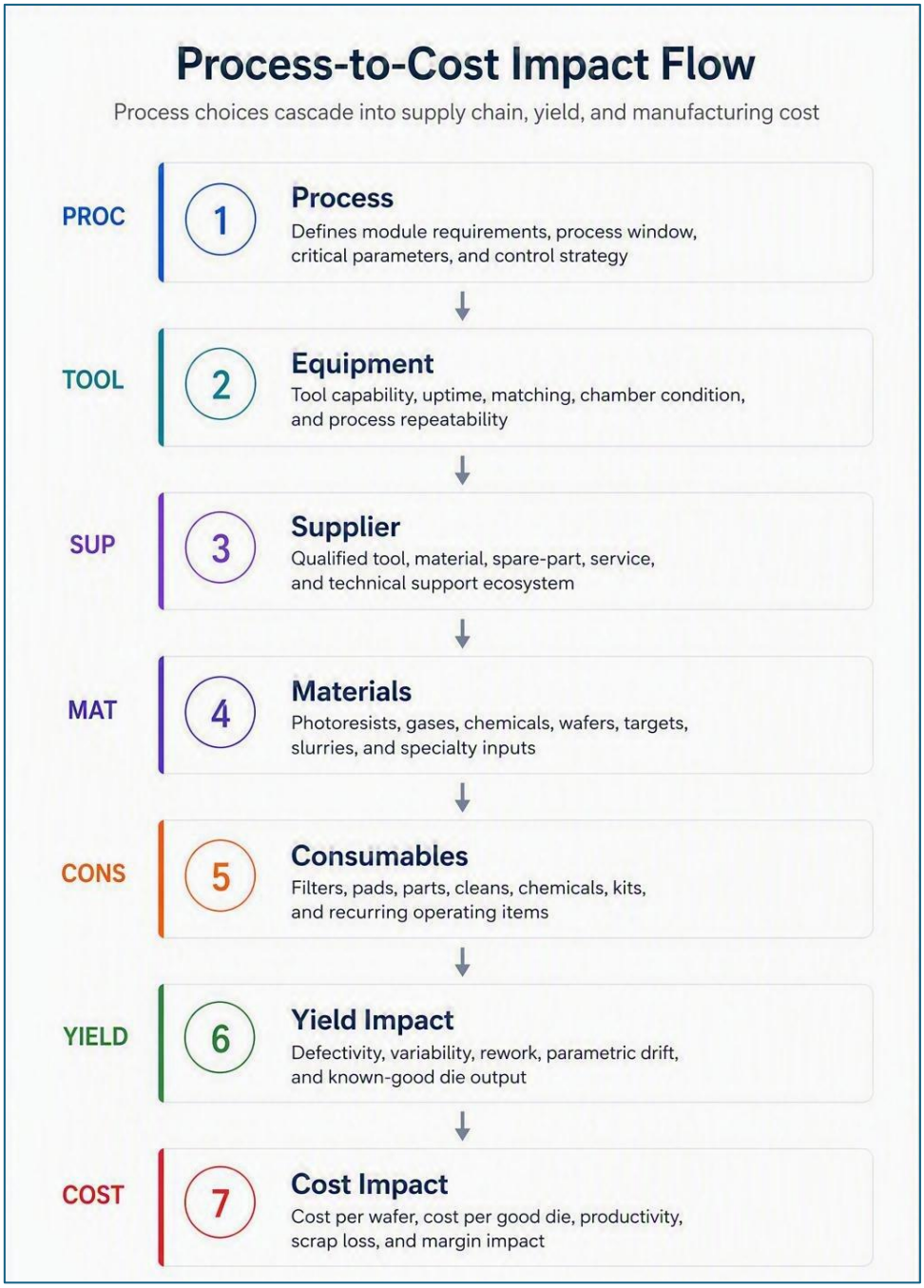


Figure 4: Process-to-cost impact flow showing how semiconductor process requirements cascade through equipment, suppliers, materials, and consumables to influence yield and total manufacturing cost.

4. Equipment bill of materials (BOM)

The equipment BOM is one of the most important components of fab economics. Knowing that a fab requires lithography, etch, deposition and CMP tools is not enough. The business case depends on:

- Tool count
- Throughput
- Availability
- Bottleneck capacity
- Utilization
- Maintenance
- Chamber configuration
- Redundancy
- Qualification requirements
- Spare-parts availability
- Tool lifetime

The correct calculation is:

Process flow → process steps → throughput requirement → tool capacity → tool count

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4.1 Main equipment categories

Module	Key tools	Typical suppliers
Lithography	ArF immersion scanner, ArF dry scanner, KrF scanner, coat/develop track	ASML, Nikon, Canon, TEL, SCREEN
Etch	Dielectric etch, conductor etch, spacer etch, hard-mask etch, strip	Lam Research, TEL, Applied Materials
Deposition	CVD, PECVD, ALD, PVD, epitaxy	Applied Materials, Lam Research, TEL, ASM
Implant	High-current, medium-current, high-energy implanters	Applied Materials, Axcelis
Thermal	Rapid Thermal Processing (RTP), furnace, spike anneal	TEL, Applied Materials, Kokusai
CMP	Polishers, post-CMP clean, slurry delivery	Applied Materials, Ebara, Lam Research
Cleans	Wet benches, single wafer cleans, megasonic cleans	SCREEN, TEL, Lam Research
Metrology	CD-SEM, overlay, films, OCD, X-ray metrology	KLA, Applied Materials, Onto, Nova

Inspection	Optical inspection, e-beam review, wafer inspection	KLA, Applied Materials, Onto
Automation	AMHS, FOUP handling, MES, APC, FDC	Daifuku, Murata, Siemens, Applied Materials, PDF Solutions

DUV lithography systems – including ArF immersion, ArF dry and KrF systems – remain important for mature and mid-mature semiconductor manufacturing where EUV is not required.

4.2 Indicative Equipment CAPEX Split

Module	Typical Share of Equipment Budget
Lithography	25-35%
Etch	15-20%
Deposition	15-20%
CMP	5-10%
Implant	5-10%
Metrology and inspection	10-15%
Automation	2-5%

Global 300mm fab equipment spending is projected by SEMI to reach US$133B in 2026, US$151B in 2027, US$155B in 2028, and US$172B in 2029, showing the scale of global equipment investment.[6]

4.3 Indicative tool-count logic

A full 40,000-WSPM fab requires sufficient tool capacity to prevent manufacturing bottlenecks. Exact counts depend on:

- Number of process steps
- Number of critical layers
- Wafer cycle time
- Tool throughput
- Tool availability
- Preventive maintenance
- Product mix
- Process duplication
- Chamber configuration
- Redundancy strategy
- Ramp requirements

Module	Tool-count logic	Indicative tools for 40k WSPM	Short reason for estimate
Lithography	Critical-layer count, scanner throughput, track capacity and utilization	25–35 scanners + 15–20 tracks	40/45 nm uses multiple DUV tool classes; high scanner throughput means fewer systems are needed than the number of lithography layers might suggest.
Etch	Etch steps per wafer, chamber productivity and recipe mix	35–50 systems	Many etch steps, but multi-chamber platforms provide high parallel capacity.
Deposition	Film-stack complexity and batch/single-wafer requirements	30–45 systems	Multiple CVD, PVD, ALD and other deposition applications require dedicated capacity.
CMP	Planarization levels, throughput and post-CMP cleaning	15–25 systems	Relatively few CMP steps and high tool throughput, with some material-specific duplication.
Implant	Implant steps, dose/energy requirements and throughput	10–18 systems	High throughput but several energy/current regimes require separate capacity.
Thermal / RTP / Furnace	Anneal, oxidation, diffusion and activation steps	25–40 systems	Numerous thermal steps, with batch furnaces providing high wafer capacity.
Wet Clean / Strip	Cleaning frequency, strip steps and throughput	15–25 systems	Cleaning occurs repeatedly throughout the flow and requires several dedicated process types.
Metrology	Sampling rate, control plan and measurement time	35–55 systems	Many measurements are required, but only a controlled sample of wafers/layers is typically measured.
Inspection	Defect-learning intensity, inspection frequency and quality requirements	10–20 systems	Inspection is concentrated on critical layers and process-control points rather than every wafer/step.

Important: *A new entrant should not minimize metrology and inspection simply to reduce initial CAPEX. During ramp, these systems are essential for:*

- *Defect identification*
- *Root-cause analysis*
- *Process-window development*
- *Excursion detection*
- *Yield learning*
- *Customer qualification*

Underinvestment can therefore reduce CAPEX while increasing the time required to achieve stable yield.

5. Materials BOM

A fab is a high-purity materials consumption engine. For a 28nm/45nm fab, materials are not simply operating inputs. Their following properties can directly affect process capability, yield and reliability:

- Purity
- Particle performance
- Metal contamination
- Chemical stability
- Lot-to-lot consistency
- Shelf life
- Packaging
- Supply reliability

5.1 Main Materials Categories

Module	Key Materials
Lithography	ArF immersion resist, ArF dry resist, KrF resist, BARC, SOC, topcoat, developer, rinse, EBR, solvents
Etch	CF_4, CHF_3, C_4F_8, SF_6, Cl_2, HBr, O_2, Ar, He, chamber clean gases
Deposition	SiH_4, TEOS, NH_3, N_2O, WF_6, Cu/Al/Ti/Ta targets, high-k precursors
CMP	Oxide slurry, tungsten slurry, copper slurry, barrier slurry, pads, conditioners, post-CMP cleans
Cleaning	HF, H_2SO_4, H_2O_2, NH_4OH, HCl, IPA, TMAH, solvents, UPW

We can describe semiconductor fabrication as relying on gases, chemicals, solvents, ultrapure water, deposition precursors, ion implantation gases, cleaning chemistries, and repeated process control steps.[10]

5.2 Annual material cost model

The manufacturer should calculate material cost as follows:

$$Annual\ material\ cost = Material\ cost\ per\ processed\ wafer \times Annual\ processed\ wafer$$

$$Annual\ processed\ wafer = 40{,}000\ WSPM \times 12 \times 85\% = \mathbf{408{,}000\ wafers\ per\ year}$$

Assume mature-node material cost:

Scenario	Material Cost / Wafer	Annual Materials Cost
Conservative	US$700	US$286M
Base	US$900	US$367M
High	US$1,100	US$449M

5.3 Representative Materials Spend Split

Category	Share of Materials Spend	Base-Case Annual Spend at US$367M
Silicon wafers	15-25%	US$55M-92M
Lithography materials	15-20%	US$55M-73M
Wet chemicals	10-15%	US$37M-55M
Gases	10-15%	US$37M-55M
CMP materials	10-15%	US$37M-55M
Deposition precursors / targets	10-20%	US$37M-73M
Masks / Pellicles related	5-15%	US$18M-55M

The next layer should include:

- Safety stock
- Supplier lead time
- Dual-source availability
- Qualification time
- Geographic concentration
- Emergency supply capability
- Change-control requirements

Key manufacturing lesson: *Materials are not commodities at 28nm/45nm. Supplier quality, metal contamination, particle control, storage conditions, shelf life, lot-to-lot variation, and change-control discipline directly affect yield.*

NEXT CHAPTER AVAILABLE IN A WEEK

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